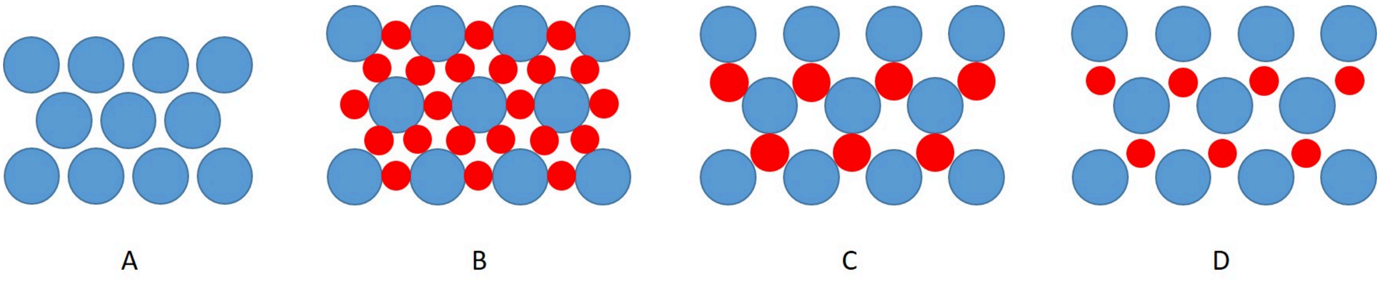
41680 F24 Intro to Advanced Materials Exam (adjustet for F25)

Der anvendes en scoringsalgoritme, som er baseret på "One best answer"

Dette betyder følgende:
Der er altid netop ét svar som er mere rigtigt end de andre
Studerende kan kun vælge ét svar per spørgsmål
Hvert rigtigt svar giver 1 point
Hvert forkert svar giver 0 point (der benyttes IKKE negative point)

The following approach to scoring responses is implemented and is based on "One best answer"
There is always only one correct answer – a response that is more correct than the rest
Students are only able to select one answer per question
Every correct answer corresponds to 1 point
Every incorrect answer corresponds to 0 points (incorrect answers do not result in subtraction of points)

The figure illustrates potential lattice planes in ionic crystals. Anions are shown in blue, cations in red, radii and distances are properly scaled.

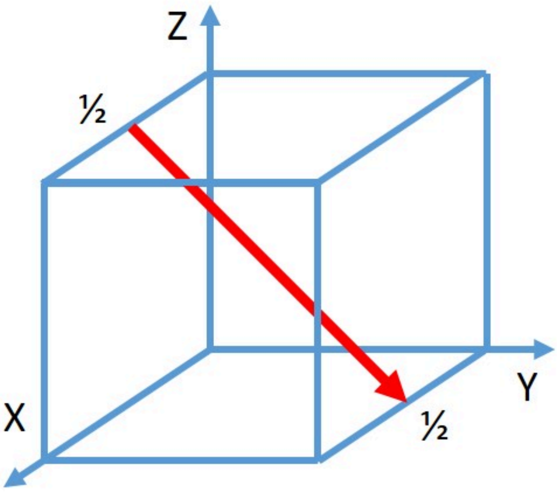


Which of the lattice planes represents a {111} plane in a stable sodium chloride structure?

Choose one answer

- A
- B
- C
- D

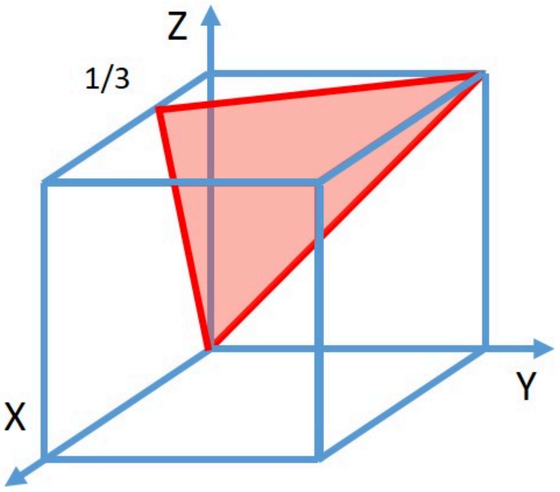
What is the proper notation of the particular crystallographic direction shown in the unit cell in the figure?



Choose one answer

- $[01\bar{1}]$
- $[12\bar{2}]$
- $[011]$
- $[122]$

What is the proper notation of the particular crystallographic plane shown in the unit cell in the figure?



Choose one answer

- (133)
- (311)
- $(13\bar{3})$
- $(31\bar{1})$

The fictive element scheelium forms a body-centered cubic lattice with a lattice constant of 316.5 pm and a density of 19.25 g/cm^3. What is the molar mass of scheelium?

Choose one answer

- ○ 45.9 g/mol
- ○ 91.9 g/mol
- ● 183.8 g/mol
- ○ 367.5 g/mol

Strontium has two allotropes: α-Sr with face-centered cubic lattice and β-Sr with body-centered cubic lattice. Which statement comparing both allotropes is correct?

Choose one answer

- [x] α-Sr has a larger lattice constant than β-Sr.
- [ ] α-Sr has a larger unoccupied volume fraction between atoms than β-Sr.
- [ ] β-Sr has larger interstitial sites than α-Sr.
- [ ] β-Sr has a larger density than α-Sr.

The table provides the electronegativities of a few elements.

Element	Electronegativity (according to Pauling)
Al	1.5
N	3.0
O	3.5
S	2.5
Si	1.8
Zr	1.4

Order the three ceramic compounds, alumina Al_2O_3, silicon nitride Si_3N_4 and zirconium disulfide ZrS_2, from the lowest to highest fraction of covalent bonding.

Choose one answer

- (selected) Al_2O_3, Si_3N_4, ZrS_2
- Al_2O_3, ZrS_2, Si_3N_4
- Si_3N_4, ZrS_2, Al_2O_3
- Si_3N_4, Al_2O_3, ZrS_2
- ZrS_2, Al_2O_3, Si_3N_4
- ZrS_2, Si_3N_4, Al_2O_3

HARD

The table lists different properties of the elements francium and bromine.

Use ionic radius not atomic raidus

Element	Molar mass / g/mol	Atomic radius / nm	Ionic radius / nm	Valence	Elektronegativity (according to Pauling)
Francium	223.0	0.260	0.194	+1	0.7
Bromine	79.9	0.115	0.196	-1	2.8

Which crystal structure and which mass density is expected for francium bromide (FrBr)?

Choose one answer

- CsCl structure with a density of about 5.5 g/cm^3
- CsCl structure with a density of about 6.2 g/cm^3
- NaCl structure with a density of about 4.8 g/cm^3
- NaCl structure with a density of about 4.3 g/cm^3

The table lists different properties of six elements, all with body-centered cubic lattice.

Element	Mo	Ba	Cr	Nb	Ta	V
Molar mass / g/mol	95.9	137.3	52.0	92.9	180.9	50.9
Density / g/cm^3	10.2	8.9	7.2	8.6	16.7	6.1
Atomic radius / pm	136	136	125	143	138	132
Electronegativity	1.8	0.9	1.6	1.6	1.5	1.6
Valence	+4	+2	+3	+5	+5	+5

Which of the elements cannot be expected to be completely miscible with molybdenum?

Choose one answer

- (●) Ba **There is a big difference in electronegativity between Mo & Ba**
- ( ) Cr
- ( ) Nb
- ( ) Ta
- ( ) V

Nickel (Ni) shall be alloyed with either gold (Au) or copper (Cu) to increase its strength. Ni and Cu are completely miscible in the solid state, while Ni and Au are completely immiscible in the solid state. The table lists some mechanical properties of the elements.

Element	Au	Cu	Ni
Density / g/cm^3	19.32	8.96	8.90
Young's modulus / GPa	78	130	208
Yield strength / MPa	20	25	59

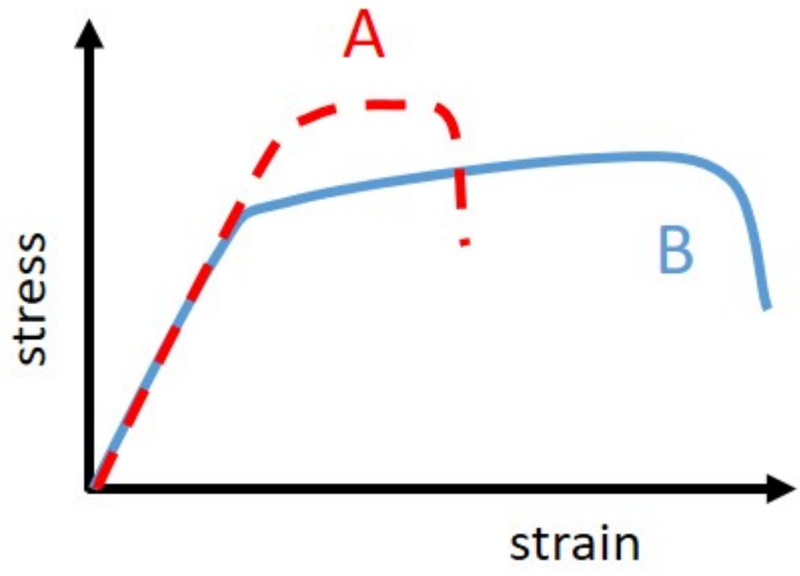
Can an alloy stronger than pure nickel be obtained by alloying Ni either with 1 wt.% Cu or with 1 wt.% Au?

Choose one answer

- [x] Yes, but only by alloying with Cu.
- [ ] Yes, but only by alloying with Au.
- [ ] Yes, in both cases, alloying with either Cu or Au.
- [ ] No, neither by alloying with Cu, nor with Au.

If miscible, the materials becomes stronger.
If Immiscible, the materials becomes weaker.

The figure shows the results of tensile tests for two specimens A and B.



Which statement is correct?

Choose one answer

- ○ Specimen B is stiffer than specimen A. **Stiffness = young's modulus**
- ○ Specimen B is more compliant than specimen A. **Compliant = (young's modulus)^(-1)**
- ○ Specimen B is stronger than specimen A. **No, max stress is lower in B**
- ● Specimen B is tougher than specimen A. **Yes, the area under the curve is bigger**

Resilience is area under the curve, but only for linearly/elastic part!

Which defects contribute to the strength of a metallic material?

Choose one answer

- ○ Dislocations and pores
- ○ Dislocations and vacancies
- ● Grain boundaries and interstitials
- ○ Grain boundaries and cracks

Help strength:
- Dislocations
- Grain boundaries
- Interstitials

Nothing to strength:
- Vacancies

Worsens strength:
- Pores

What is the microstructural reason for work-hardening?

Choose one answer

- Formation of interstitials
- Accumulation of dislocations
- Growth of grains
- Precipitation of second phases

Work-hardening (also called strain hardening or cold working) is the process of increasing the strength and hardness of a metal by plastically deforming it — without adding heat.

Complete the statement: increasing the dislocation density in pure gold will increase …

Choose one answer

- ... its mass density.
- … its Young’s modulus.
- … its resilience.
- … its ductility. **decrease in ductility with dislocations**

Compare four specimens of pure nickel, which differ in grain size and dislocation density according to the table.

Specimen	Grain size / µm	Dislocation density / m^{-2}
A	15	$3 \cdot 10^{14}$
B	15	$2 \cdot 10^{15}$
C	50	$3 \cdot 10^{14}$
D	50	$2 \cdot 10^{15}$

Which of the four specimens is hardest?

Choose one answer

- ○ A
- ● B **smallest grain size, highest dislocation density**
- ○ C
- ○ D

Complete the statement: The strength of a ceramic decreases …

Choose one answer

- ○ … with decreasing porosity. **increases strenght, because they will gaps**
- ○ … with decreasing temperature. **more brittle, more strenght**
- ○ … with decreasing grain size. **decreasing grain size, increases strength**
- ● … with decreasing density. **decreasing density -> more pores -> lower strenght**

Compare two batches of polyethylene with same degree of polymerization. What could be a valid reason for a batch A having a higher melting temperature than a batch B?

Choose one answer

- ○ Batch A is atactic, while batch B is isotactic.
- ○ Batch A is isotactic, while batch B is atactic.
- ○ Batch A has more side branches than batch B.
- ● Batch A has less side branches than batch B.

Isotactic = All side groups are on the same side of the polymer chain.

Atactic = Side groups are arranged randomly along the chain

Not relevant for this case

Which polymer has the longest straightened chain length?

Choose one answer

- [x] Polybutadiene with a degree of polymerization of 300.
- [ ] Polypropylene with a degree of polymerization of 380.
- [ ] Polyethylene with molar mass of 14 kg /mol.
- [ ] Polyvinylchloride with molar mass of 25 kg /mol.

What is characteristic for the mechanical behavior of semi-crystalline polymers?

Choose one answer

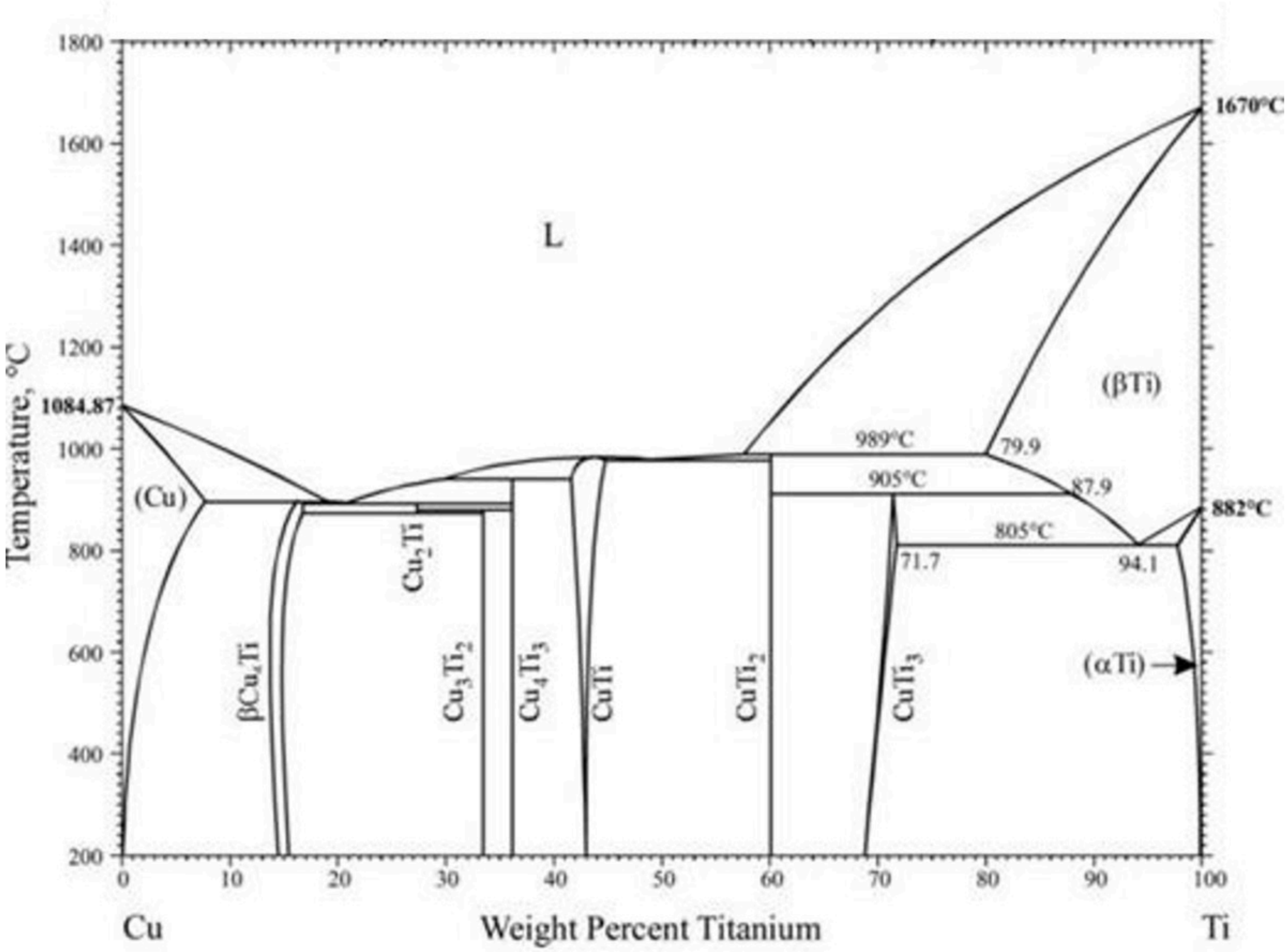
- They have an extremely low Young’s modulus.
- They deform in a viscous manner without necking.
- They work-harden after an initial decrease in flow stress.
- They fracture brittle without plastic deformation.

Why are elastomers extremely compliant?

Choose one answer

- ○ The covalent bonds along the polymer chain are weak.
- ○ The van der Waals interaction between polymer chains is too strong.
- ○ There are only a few hydrogen bonds between the side groups.
- ● Their compliance is not determined by the bond strength.

The figure shows the phase diagram of the binary system copper-titanium (Cu-Ti). No changes occur between 200 °C and room temperature.

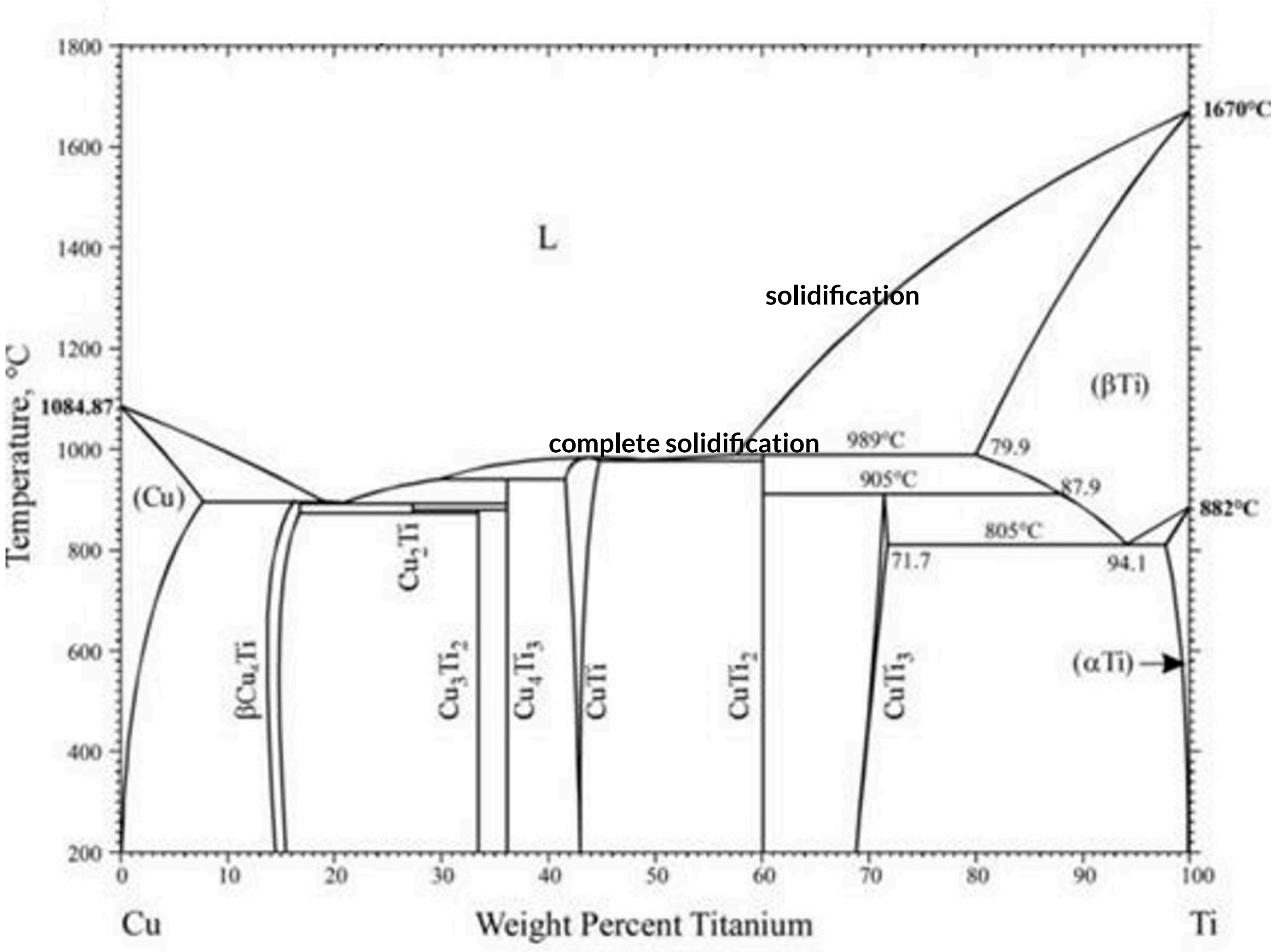


What is the maximal solubility of copper in solid titanium?

Choose one answer

- ○ About 2 wt.%
- ○ About 8 wt.%
- ● About 20 wt.%
- ○ About 80 wt.%
- ○ About 92 wt.%
- ○ About 98 wt.%

The figure shows the phase diagram of the binary system copper-titanium (Cu-Ti). No changes occur between 200 °C and room temperature.

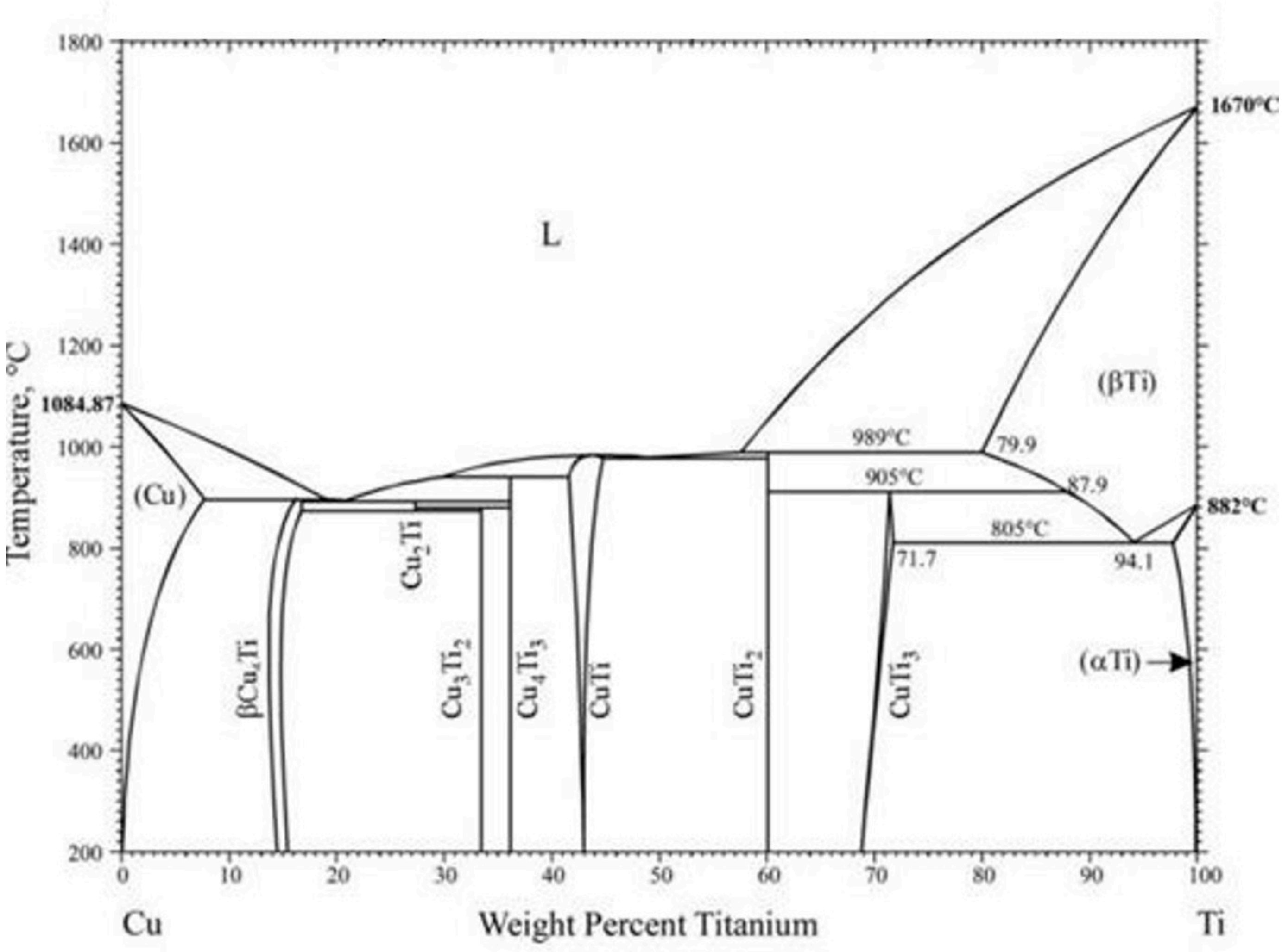


A melt of a binary copper-titanium alloy with 75 wt.% Ti and 25 wt.% Cu is cooled slowly from 1800 °C to room temperature. At which temperature does solidification finish?

Choose one answer

- 1360 °C
- 989 °C
- 905 °C
- 805 °C

The figure shows the phase diagram of the binary system copper-titanium (Cu-Ti). No changes occur between 200 °C and room temperature.

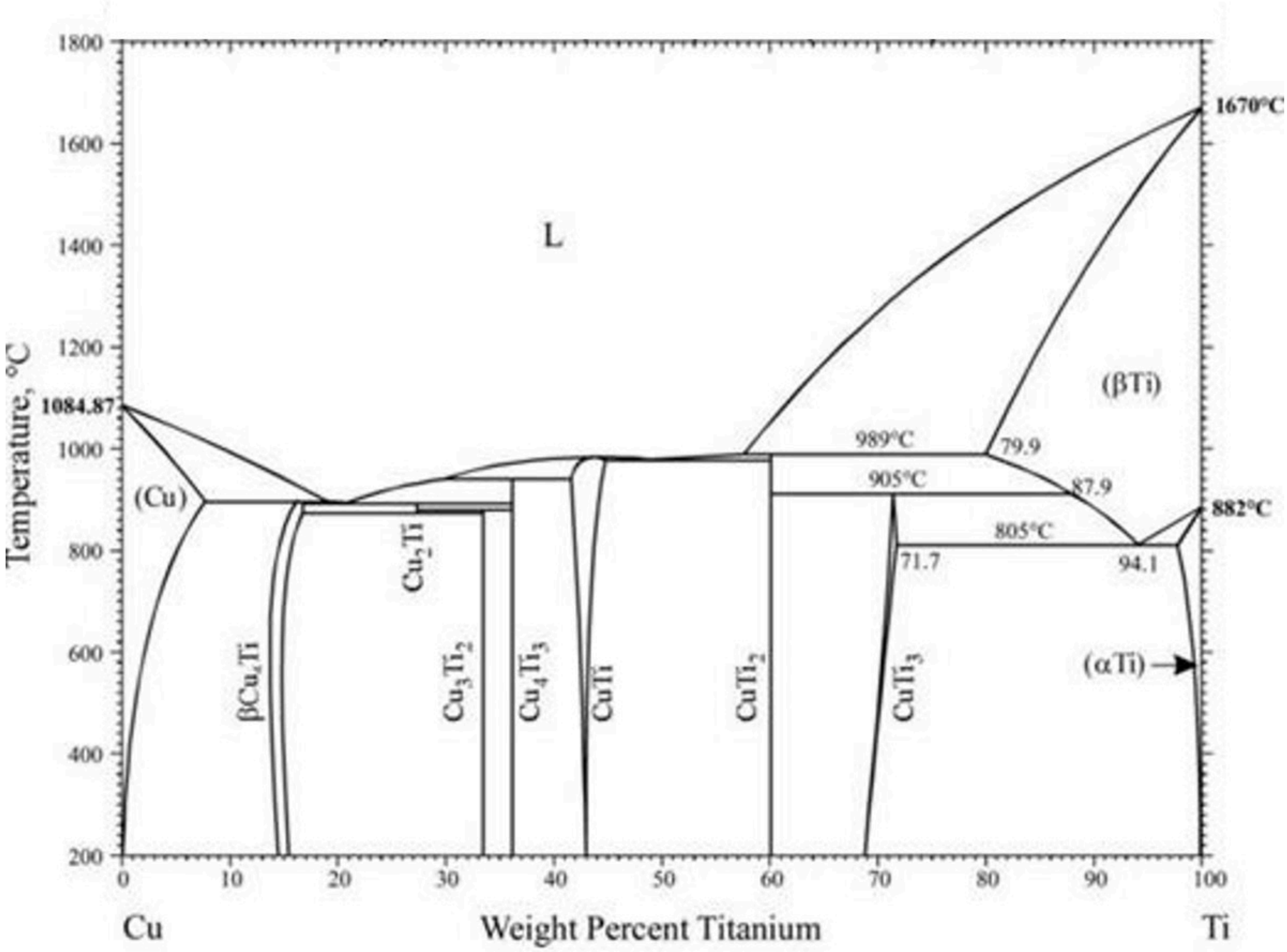


Where are copper atoms expected to be found at 850 °C for a binary copper-titanium alloy with 75 wt.% Ti and 25 wt.% Cu?

Choose one answer

- On regular lattice sites in $CuTi_3$.
- On interstitial sites in (αTi).
- On regular lattice sites in (αTi).
- On interstitial sites in (βTi).

The figure shows the phase diagram of the binary system copper-titanium (Cu-Ti). No changes occur between 200 °C and room temperature.



A melt of a binary copper-titanium alloy with 75 wt.% Ti and 25 wt.% Cu is cooled slowly from 1800 °C to room temperature. How large is the mass of (αTi) at room temperature formed from a melt with a mass of 8 kg?

Choose one answer

- 0 kg
- About 1 kg
- About 1.5 kg
- About 6.5 kg
- About 7 kg
- 8 kg

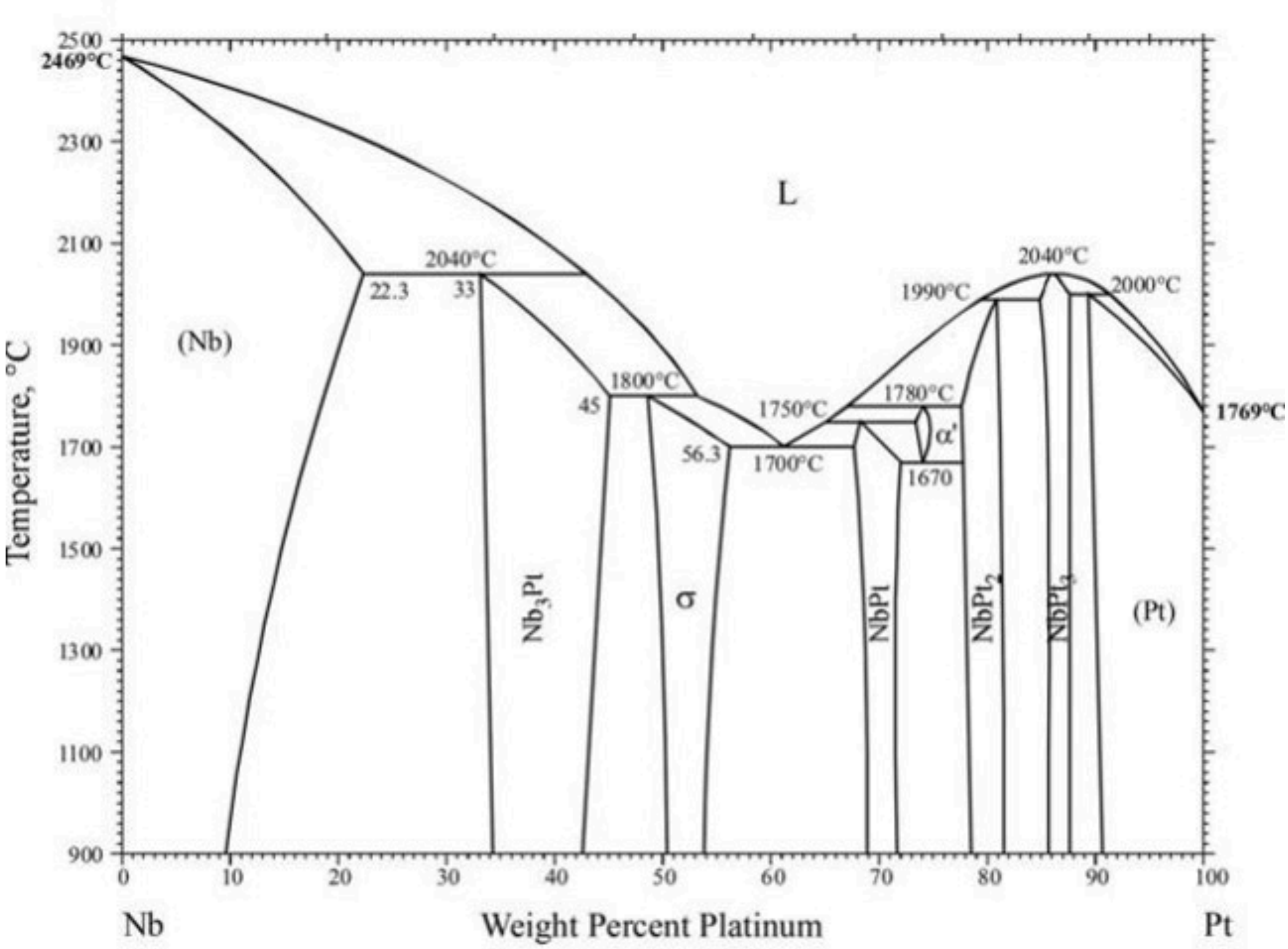
Use lever rule:
CuTi3 = 69%
alphaTi = 100%

we want 75%.

75% = 69%*(x-1) + 100*(x), solve for x

answer: x*8

The figure shows the phase diagram of the binary system niobium-platinum (Nb-Pt). No changes occur between 900 °C and room temperature.

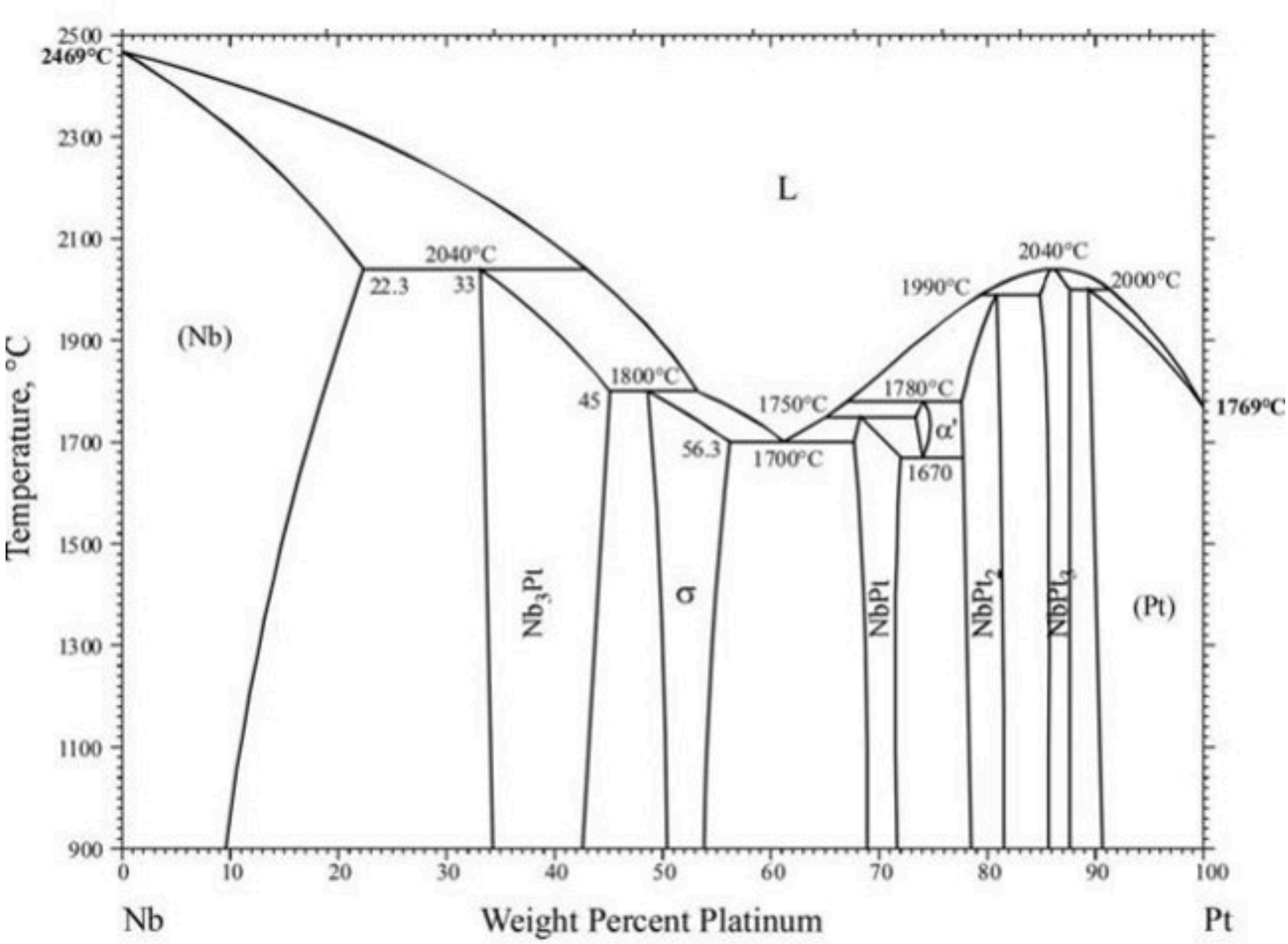


How many peritectic transformations can be identified in the diagram?

Choose one answer

- 0
- 1
- 2
- 5
- 6
- 7

The figure shows the phase diagram of the binary system niobium-platinum (Nb-Pt). No changes occur between 900 °C and room temperature.

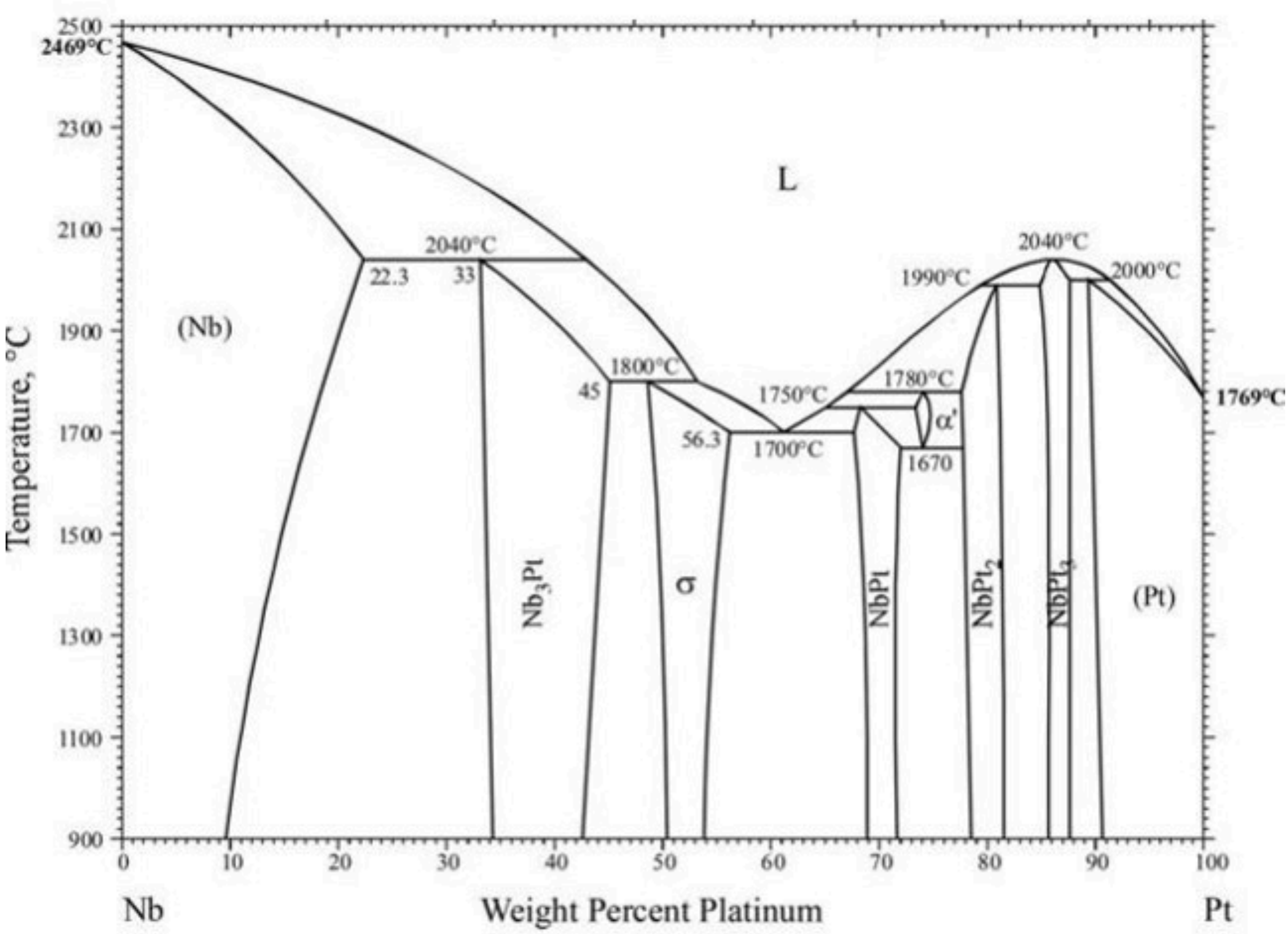


A melt of a binary niobium-platinum alloy with 38 wt.% Pt is cooled slowly from 2600 °C to room temperature. What is the first solid phase formed during solidification?

Choose one answer

- (Nb)
- (Nb) + L
- Nb_3Pt
- Nb_3Pt + L
- σ
- σ + L

The figure shows the phase diagram of the binary system niobium-platinum (Nb-Pt). No changes occur between 900 °C and room temperature.

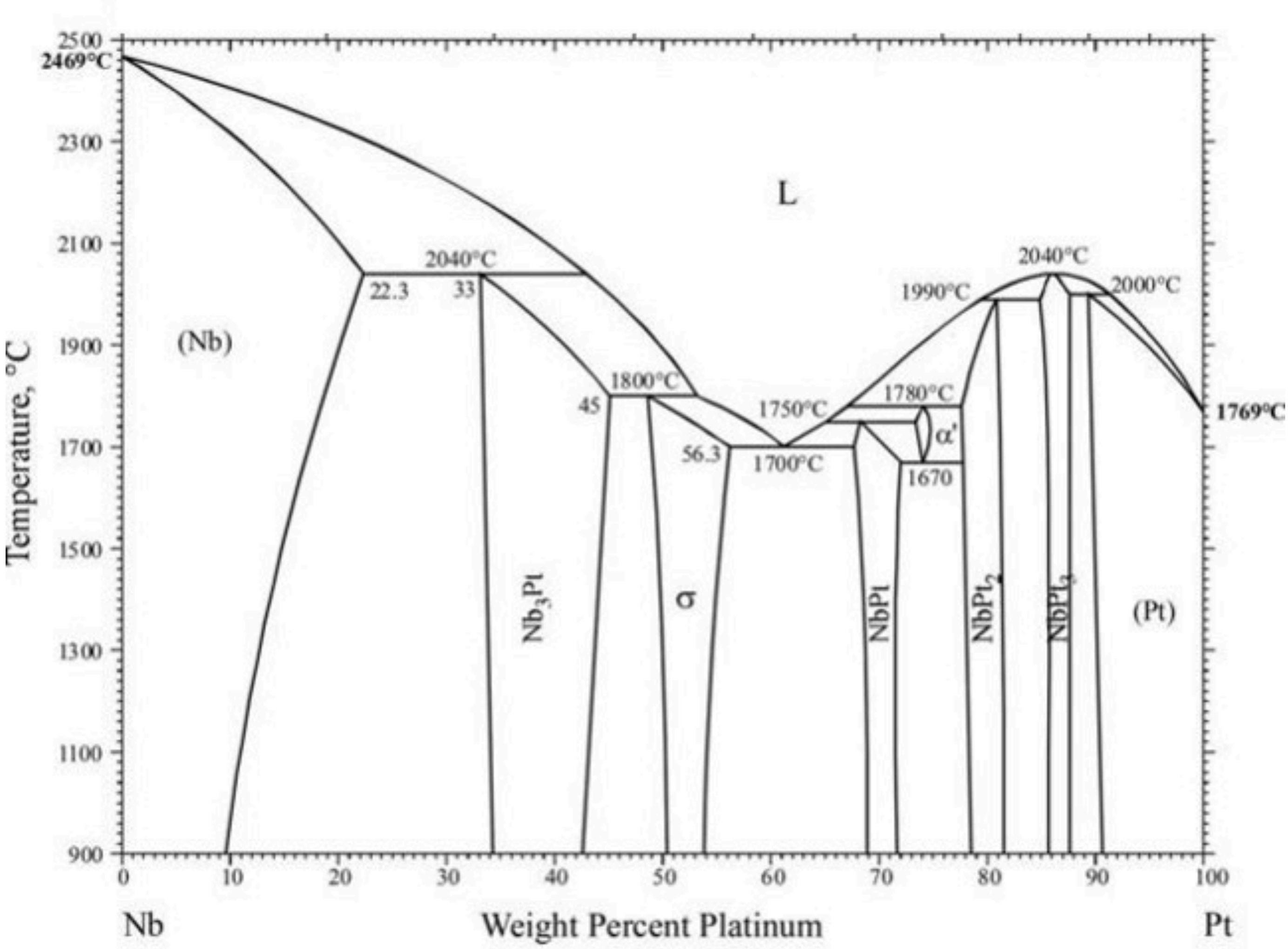


A melt of a binary niobium-platinum alloy with 38 wt.% Pt is cooled slowly from 2600 °C to room temperature. What is the largest content of platinum in the melt during solidification?

Choose one answer

- Exactly 38 wt.%
- About 43 wt.%
- About 47 wt.%
- About 53 wt.%
- About 61 wt.%

The figure shows the phase diagram of the binary system niobium-platinum (Nb-Pt). No changes occur between 900 °C and room temperature.

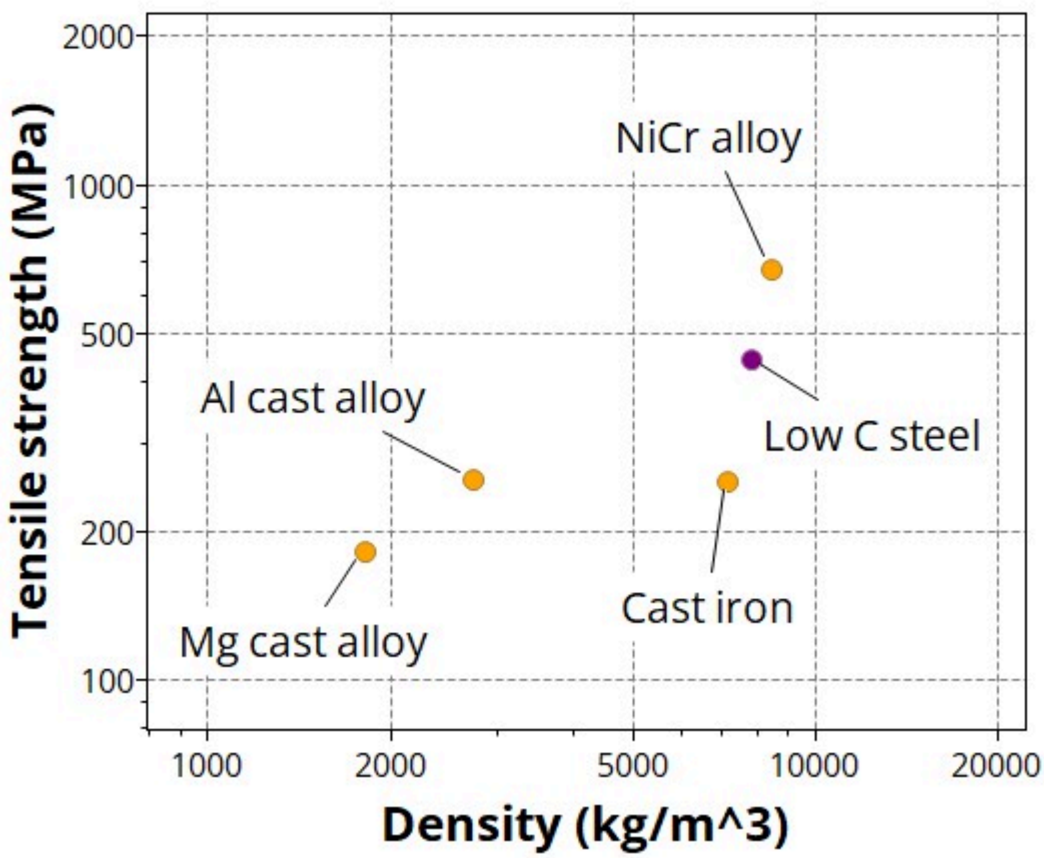


How many phases can co-exist in thermodynamic equilibrium in the microstructure of a binary niobium-platinum alloy at 1670 °C?

Choose one answer

- Not more than two phases.
- Maximal three phases.
- Maximal four phases.
- More than four phases.

A lighter beam shall replace a rectangular supporting beam of given length presently made of low carbon steel. The materials property map presents the properties of four suggested alternatives.

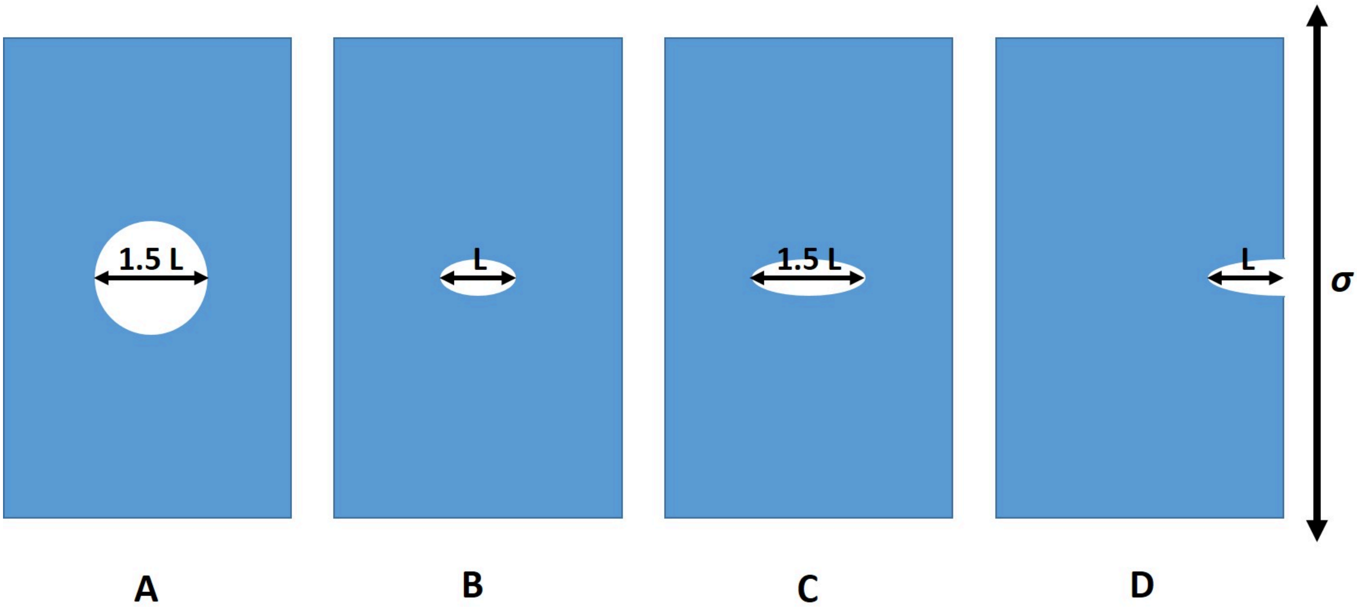


Which materials can be used to replace low carbon steel with a lighter beam still carrying the same tensile load without fracture?

Choose one answer

- ○ Only the NiCr alloy.
- ○ Only the Mg cast alloy.
- ○ Only two of the cast alloys (Mg cast alloy and Al cast alloy).
- ○ All three cast alloys (Mg cast alloy, Al cast alloy and cast iron).
- ● Three alloys (Mg cast alloy, Al cast alloy and NiCr alloy).

Cracks (as well as holes and notches) impede the mechanical properties of materials. The graph illustrates four different idealized crack configurations in a metallic sheet.

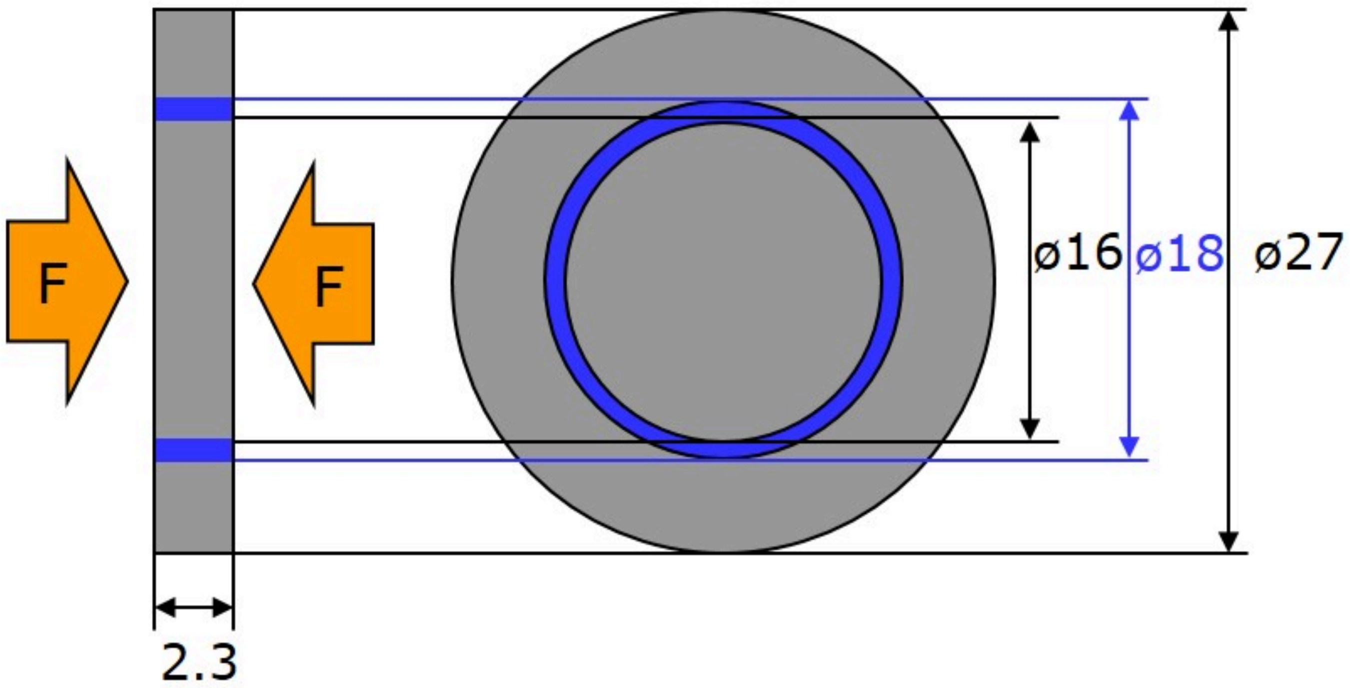


While the cracks differ in length and location, assume the same shape of the crack tip for configurations B, C, D for comparison. Which configuration is the most likely one to fail, when loaded in tension as indicated?

Choose one answer

- A
- B
- C
- D

Modern coins contain a polymer ring (shown in blue) between a metal disc and metal ring as shown in the technical drawing.



The coin is 2.3 mm in height and has a diameter of 27 mm. A copper-nickel alloy with a Young's modulus of 145 GPa is used for both, the outer metal ring and the inner disc with diameter of 16 mm. The polymer ring is 1 mm thick and has a Young's modulus of 1 GPa. What is the Young's modulus of the composite, when the coin is compressed along the thickness direction (as shown in the figure)?

Choose one answer

- Larger than 133 GPa.
- Larger than 92 GPa, but at most 133 GPa.
- Larger than 55 GPa, but at most 92 GPa.
- Larger than 11 GPa, but at most 55 GPa.
- At most 11 GPa.

A wire subjected to a voltage drop of 1 mV is observed to transport a current of 340 mA. The wire has a cylindrical cross section with a diameter of 1 mm and length of 10 cm. The table lists the electrical conductivities of several metals.

Metal	Electrical conductivity / $(\Omega m)^{-1}$
Gold	$4.3 \cdot 10^7$
Aluminium	$3.8 \cdot 10^7$
Platinum	$0.94 \cdot 10^7$
Stainless steel	$0.2 \cdot 10^7$

Which material is consistent with the observation?

Choose one answer

- Gold
- Aluminium
- Platinum
- Stainless Steel

An aluminium wire has a cross section of $1.6 \cdot 10^{-6}$ m^2 and carries an electrical current of 2 A. Each Al atom contributes one free electron. The molar mass of Al is 26.98 g/mol, while its density is 2.7 g/cm^3. What is the drift speed of the electrons in the wire?

Choose one answer

- 0.13 mm/s
- 27 cm/s
- 37 m/s
- 5.1 km/s

Text

What type of carriers are active in conducting an electrical current in the semiconductor GaAs at room temperature?

Choose one answer

- Solely electrons in the conduction band.
- Solely holes in the conduction band.
- Electrons in the valence band and holes in the conduction band.
- Electrons in the conduction band and holes in the valence band.

Pure nickel and silicon glass have widely different thermal conductivities. What is the major reason for this difference?

Choose one answer

- ○ Ni is denser than silicon glass.
- ● Ni has a crystalline lattice, which can carry phonons, silicon glass not.
- ○ Ni is magnetic, silicon glass not.
- ○ Ni has free holes that acts as carriers, silicon glass not.

In a very simplified model, an airplane is considered as cylinder with a total surface area of 500 m^2. The wall has a thickness of 1 cm everywhere and shall be made of a material with a thermal conductivity of 0.2 W/(m K). The temperature inside the airplane is 25 °C, outside it is -50 °C. What is the total energy loss during a 10 h flight assuming a steady state heat flow?

Choose one answer

- ○ 340 kJ
- ○ 12 MJ
- ○ 82 MJ
- ● 27 GJ

Consider simulating the temperature distribution in a rod of metal as a 1D heat transfer problem using the heat equation. The rod is 1 m long. The two ends of the rod are defined at $x = 0$ m (point A) and $x = 1$ m (point B). The initial temperature of the rod is 20 °C. The boundary condition at point A is $u = 100$ °C, the boundary condition at point B is $\partial u/\partial n = 0$. In the steady state solution, what is the temperature at $x = 0.5$?

Choose one answer

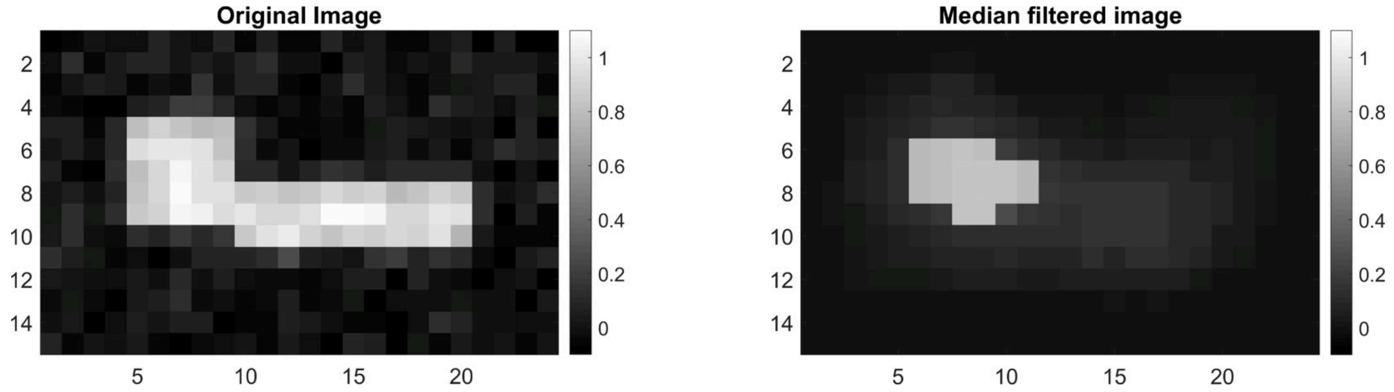
- 20 °C
- 50 °C
- 60 °C
- 100 °C
- A steady state solution does not exist.

Consider a 3D image of a material consisting of two phases A and B. The image has dimensions 200 x 300 x 400 voxels and a voxel size of 50 µm x 50 µm x 50 µm. Through 3D image analysis the phase fraction of phase A is quantified to be 0.2. What is the volume of phase A in the 3D image?

Choose one answer

- 24 μm^3
- 600 μm^3
- 3000 μm^3
- 24 mm^3
- 600 mm^3
- 3000 mm^3

Consider the original image below to the left. The result of median filtering the original image is shown to the right.

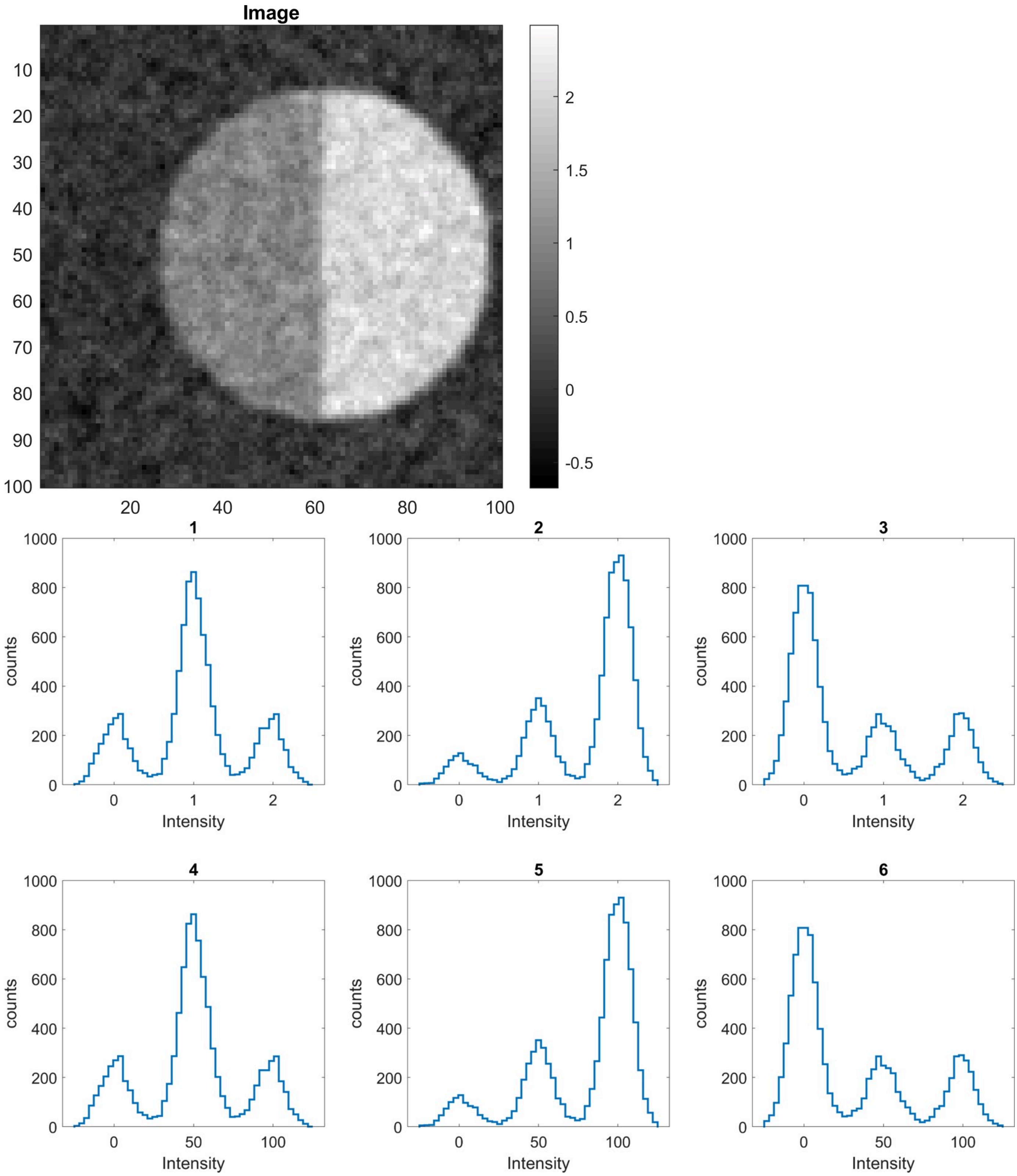


Which was the size of the median filter used?

Choose one answer

- 3 x 3 median filter
- 5 x 5 median filter
- 7 x 7 median filter
- 9 x 9 median filter

Consider the image below consisting of 100 x 100 pixels and the suggested intensity histograms.



Which of the shown intensity histograms correspond to the image? Note that the horizontal axis has not the same scale for all histograms.

Choose one answer

- 1

- 2
- 3
- 4
- 5
- 6

We have implemented a solver for the 2D heat equation

$$\frac{\partial u}{\partial t} = \alpha \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)$$

using an iterative finite difference method. What would be an appropriate stopping criterion to make sure that we stop as soon as we have reached a steady state solution?

Choose one answer

- ○ Stop when u is close to zero everywhere.
- ● Stop when $\frac{\partial u}{\partial t}$ is close to zero everywhere.
- ○ Stop when $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y}$ is close to zero everywhere.
- ○ Stop after 10000 iterations
- ○ Stop after 100 iterations